THE p-ADIC LOGARITHM FROM MULTIPLICATION TO ADDITION

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Abstract. The p-adic logarithm gives an explicit comparison between multiplicative principal units and an additive ideal. We prove convergence, exact valuation estimates, the formal identities and their legitimate evaluation, and the topological group isomorphism with inverse given by the exponential. The prime 2 is treated separately. Exact computations modulo powers of 5 illustrate how denominators determine truncation precision. We then construct Teichmüller representatives and a normalized logarithm on $\mathbb{Q}_p^\times$, identify its kernel, and explain why none of these maps is a ring homomorphism. The final discussion specifies the limited connection with the log-link of IUT. Six exercises have complete solutions.

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Date: October 13, 2026.

1. Valuations, congruences, and convergent series

Fix a prime p. For $a \in \mathbb{Q}^\times$, write $a = p^k m/n$ with integers m, n not divisible by p, and put $v_p(a) = k$, $v_p(0) = +\infty$. The associated absolute value is $|a|_p = p^{-v_p(a)}$ for $a \neq 0$, and $|0|_p = 0$. Multiplication adds valuations. Addition satisfies

$$v_p(a+b) \geq \min\{v_p(a), v_p(b)\}, \qquad |a+b|_p \leq \max\{|a|_p, |b|_p\}. \tag{1.1}$$

If the two valuations are unequal, the first inequality is an equality: factoring out the summand of smaller valuation leaves a factor congruent to 1 modulo p.

The field $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for this absolute value. Its nonzero elements still have integer valuations: sufficiently accurate rational approximations to a nonzero element have a common valuation, by the unequal-valuation assertion above. Define

$$\mathbb{Z}_p = \{a \in \mathbb{Q}_p : v_p(a) \geq 0\}, \qquad \mathbb{Z}_p^\times = \{a \in \mathbb{Q}_p : v_p(a) = 0\}.$$

The ring and unit assertions follow from (1.1) and $v_p(ab) = v_p(a) + v_p(b)$. We assume this standard construction of the completed field; we will prove the series facts used below.

For $a, b \in \mathbb{Q}_p$, the notation $a \equiv b \pmod{p^N}$ means $a - b \in p^N\mathbb{Z}_p$, equivalently $v_p(a-b) \geq N$. In particular, it is meaningful for rational fractions, including those whose denominators contain powers of p. One must first cancel those powers before using a modular inverse. Reduction gives $\mathbb{Z}_p/p^N\mathbb{Z}_p \simeq \mathbb{Z}/p^N\mathbb{Z}$. For completeness, any $x \in \mathbb{Z}_p$ can be approximated modulo p^N by a rational a/b with $p \nmid b$. Choose an integer inverse of b modulo p^N to obtain an integer representative for x. The kernel of this reduction is exactly $p^N\mathbb{Z}_p$.

Lemma 1.1 (Series criterion)**.** *In $\mathbb{Q}_p$, a series $\sum_{n\geq 0} a_n$ converges if and only if $a_n \to 0$. If every term of a convergent tail has valuation at least N, the sum of that tail also has valuation at least N.*

Proof. Convergence implies $a_n = S_n - S_{n-1} \to 0$, where S_n are partial sums. Conversely, if $a_n \to 0$, then for every N there is an index M such that $v_p(a_n) \geq N$ for $n \geq M$. Every finite sum of such terms has valuation at least N by (1.1); thus the partial sums are Cauchy and converge by completeness. The ball $p^N\mathbb{Z}_p$ is closed, so the same bound holds for the limiting tail. □

The same reasoning justifies a useful rearrangement principle. A family of terms may be summed in any order if, for every N, only finitely many terms have valuation less than N. Indeed, modulo p^N all the remaining terms are zero, so every ordering yields the same finite sum at that precision. This is the criterion we will use for products and substitutions of power series; a formal identity alone does not justify evaluation.

2. The logarithm and its convergence domain

Definition 2.1. For $x \in p\mathbb{Z}_p$, define

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}. \tag{2.1}$$

The group $U_r = 1 + p^r\mathbb{Z}_p$ is the *rth principal unit group*, for integers $r \geq 1$.

The series converges. For nonzero x and $s = v_p(x) \geq 1$,

$$v_p(x^n/n) = ns - v_p(n) \geq ns - \lfloor \log_p n \rfloor \longrightarrow +\infty. \tag{2.2}$$

Here $p^{v_p(n)} \leq n$ is the ordinary integer inequality behind the bound. At $x = 0$ the sum is zero. Within $\mathbb{Q}_p$, the series does not converge at a nonzero x with $v_p(x) \leq 0$: the terms

with $n = p^k$ have valuation $p^k v_p(x) - k$, which does not tend to $+\infty$. Thus the exact convergence domain in the variable x is $p\mathbb{Z}_p$.

The set U_r is a multiplicative group. For $x, y \in p^r\mathbb{Z}_p$,

$$(1+x)(1+y) = 1 + (x + y + xy) \in U_r.$$

The geometric series $\sum_{n\geq 0}(-x)^n$ converges and is the inverse of $1+x$, since its finite partial sums have product with $1+x$ tending to 1. Its difference from 1 lies in $p^r\mathbb{Z}_p$, so the inverse also belongs to U_r. By contrast, U_r is not a subring with the inherited operations: it does not contain 0 and is not closed under addition, since $1 + 1 = 2 \notin U_r$ for any prime p.

3. Denominator estimates and exact first-term control

The denominators in (2.1) cannot be ignored. The following estimates control both logarithms and exponentials.

Lemma 3.1. *For every integer $n \geq 1$,*

$$v_p(n) \leq \frac{n-1}{p-1}, \qquad v_p(n!) = \sum_{j\geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor = \frac{n - s_p(n)}{p-1} \leq \frac{n-1}{p-1}, \tag{3.1}$$

where $s_p(n)$ is the sum of the base-p digits of n.

Proof. If $k = v_p(n)$, then $n \geq p^k \geq 1 + k(p-1)$, the last inequality following from the binomial expansion $(1 + (p-1))^k$. This proves the first bound. In $n!$, each multiple of p contributes at least one factor p, each multiple of p^2 contributes another, and so on; counting these gives the floor sum. Write $n = \sum_{j=0}^{d} a_j p^j$. Then

$$\sum_{k\geq 1} \left\lfloor n/p^k \right\rfloor = \sum_{j\geq 1} a_j (1 + p + \cdots + p^{j-1}) = \frac{\sum_j a_j p^j - \sum_j a_j}{p-1}.$$

Since $n \geq 1$, its digit sum is at least one, proving the last bound. □

Set

$$r_0 = \begin{cases} 1, & p \text{ odd}, \\ 2, & p = 2. \end{cases} \tag{3.2}$$

The key inequality is $r_0 > 1/(p-1)$.

Proposition 3.2 (Exact valuations)**.** *If $x \neq 0$ and $s = v_p(x) \geq r_0$, then*

$$v_p(\log(1+x)) = s, \qquad v_p(\exp(x) - 1) = s, \quad \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}. \tag{3.3}$$

Both series converge on these domains, and their terms of degree at least two have valuation strictly greater than s.

Proof. By Lemma 3.1, for $n \geq 2$ each of the valuations $v_p(x^n/n)$ and $v_p(x^n/n!)$ is at least

$$ns - \frac{n-1}{p-1} = s + (n-1)\left(s - \frac{1}{p-1}\right) > s.$$

The lower bound tends to infinity, proving convergence. Since valuations are integers, all terms after the linear term lie in $p^{s+1}\mathbb{Z}_p$. The linear term x has valuation exactly s, so adding the remaining tail cannot change that valuation. □

For example, if $p = 5$ and $v_p(x) = 1$, then $\log(1+x) \equiv x \pmod{25}$ and $\exp(x) \equiv 1+x \pmod{25}$. This also explains the restriction at 2: for $x = 2$, the first two logarithmic terms 2 and $-2^2/2 = -2$ already have the same valuation. The first term need not dominate on $1 + 2\mathbb{Z}_2$.

4. Formal identities and why they can be evaluated

Work first in the formal power series ring $\mathbb{Q}[[X]]$. Put

$$E(X) = \sum_{n\geq 0} \frac{X^n}{n!}, \qquad L(X) = \sum_{n\geq 1} (-1)^{n+1} \frac{X^n}{n}.$$

Formal composition is defined when the inner series has zero constant term: each coefficient of a specified degree is then a finite sum.

Lemma 4.1 (Formal identities)**.** *In the indicated formal power series rings,*

$$E(X+Y) = E(X)E(Y), \qquad L(E(X)-1) = X, \qquad E(L(X)) = 1+X. \tag{4.1}$$

Proof. The coefficient of X^aY^b in $E(X+Y)$ is $\binom{a+b}{a}/(a+b)! = 1/(a!b!)$, also its coefficient in $E(X)E(Y)$. Formal differentiation gives $E' = E$ and $L' = 1/(1+X)$. By the formal chain rule,

$$\frac{d}{dX} L(E(X)-1) = \frac{E'(X)}{E(X)} = 1.$$

The series has constant term zero, so it equals X. For the last identity let $G(X) = E(L(X))$. Then $G(0) = 1$ and $(1+X)G' = G$. If $G = \sum g_n X^n$, coefficient comparison gives $g_1 = g_0 = 1$ and $(n+1)g_{n+1} = (1-n)g_n$ for $n \geq 1$. Thus $g_2 = 0$ and then every g_n for $n \geq 2$ is zero. This proves $G = 1 + X$. The formal chain rule itself follows by differentiating powers and comparing coefficients; only finite sums occur in each degree. □

Proposition 4.2 (Legitimate substitution)**.** *If $x, y \in p^{r_0}\mathbb{Z}_p$, the formal identities in* (4.1) *may be evaluated at x, y. Consequently*

$$\exp(x+y) = \exp(x)\exp(y), \qquad \log(\exp(x)) = x, \qquad \exp(\log(1+x)) = 1+x. \tag{4.2}$$

Proof. Choose an integer $r \geq r_0$ such that $x, y \in p^r\mathbb{Z}_p$; $r = r_0$ always suffices. In the product of exponentials a term of total degree $m = a + b$ has valuation at least $rm - m/(p-1)$. There are finitely many such terms in each degree, and this bound tends to infinity. The rearrangement principle after Lemma 1.1 justifies multiplication and grouping by degree, proving the first identity.

For $L(E(x)-1)$, expand the kth outer term into products of k inner terms of degrees $n_1, \ldots, n_k \geq 1$, of total degree $m = \sum n_j$. Its denominator has valuation

$$v_p(k) + \sum_{j=1}^{k} v_p(n_j!) \leq \frac{k-1+m-k}{p-1} = \frac{m-1}{p-1}.$$

For $E(L(x))$ the corresponding denominator estimate is

$$v_p(k!) + \sum_{j=1}^{k} v_p(n_j) \leq \frac{k-1+m-k}{p-1} = \frac{m-1}{p-1}.$$

In either case every expanded term of degree m has valuation at least

$$rm - \frac{m-1}{p-1},$$

which tends to infinity. For a fixed m there are only finitely many ordered compositions $m = n_1 + \cdots + n_k$, so only finitely many terms fail any prescribed valuation bound. This

proves that the expanded families can be grouped either by outer degree or by total degree. Grouping by total degree permits use of Lemma 4.1. Grouping by outer degree gives the actual convergent compositions: Proposition 3.2 ensures $\exp(x) - 1 \in p^r\mathbb{Z}_p$ and $\log(1+x) \in p^r\mathbb{Z}_p$. This proves both inverse identities without an unjustified interchange of infinite sums. □

5. The principal-unit group isomorphism

Theorem 5.1. *For every integer $r \geq r_0$, logarithm and exponential are mutually inverse topological group isomorphisms*

$$\log : (U_r, \times) \xrightarrow{\sim} (p^r\mathbb{Z}_p, +), \qquad \exp : (p^r\mathbb{Z}_p, +) \xrightarrow{\sim} (U_r, \times). \tag{5.1}$$

In particular, for odd p, $\log : (1 + p\mathbb{Z}_p, \times) \simeq (p\mathbb{Z}_p, +)$. For $p = 2$, the corresponding statement holds on $1 + 4\mathbb{Z}_2$ and $4\mathbb{Z}_2$. For $u, v \in U_r$,

$$v_p(\log u - \log v) = v_p(u - v), \tag{5.2}$$

including $u = v$ with both sides $+\infty$.

Proof. Proposition 3.2 shows that the two maps take values in the stated sets. Proposition 4.2 makes them mutually inverse. Given $u, v \in U_r$, write $u = \exp(a)$, $v = \exp(b)$ with $a, b \in p^r\mathbb{Z}_p$. Then

$$uv = \exp(a + b), \qquad \log(uv) = a + b = \log u + \log v.$$

Thus both maps preserve the specified group operations. For the valuation equality, $u/v \in U_r$ and $\log u - \log v = \log(u/v)$. When $u \neq v$, Proposition 3.2 gives

$$v_p(\log(u/v)) = v_p(u/v - 1) = v_p(u - v),$$

because v is a unit. The equality shows that log is an isometry for the metrics induced from $\mathbb{Q}_p$, and hence continuous. Its inverse is also an isometry, establishing the topological assertion. □

Corollary 5.2 (Finite quotients)**.** *If $N > r \geq r_0$, logarithm induces a group isomorphism*

$$U_r/U_N \simeq p^r\mathbb{Z}_p/p^N\mathbb{Z}_p \simeq \mathbb{Z}/p^{N-r}\mathbb{Z}. \tag{5.3}$$

For $u, v \in U_r$, $u \equiv v \pmod{p^N}$ if and only if $\log u \equiv \log v \pmod{p^N}$.

Proof. Theorem 5.1 maps U_N exactly onto $p^N\mathbb{Z}_p$, so its isomorphism descends to the quotients. The last assertion follows either from (5.2) or from $u/v \in U_N$ if and only if $u - v \in p^N\mathbb{Z}_p$. □

The isomorphism turns integral powers into multiples: $\log(u^m) = m \log u$ for every $m \in \mathbb{Z}$. It also gives a well-defined p-adic power operation

$$u^a := \exp(a \log u), \qquad a \in \mathbb{Z}_p, \quad u \in U_r.$$

This agrees with integer powers, is continuous in a, and satisfies $u^{a+b} = u^a u^b$ and $(u^a)^b = u^{ab}$ by (4.2). These are statements about the principal unit group, not an unrestricted exponentiation operation on $\mathbb{Q}_p^\times$.

6. Exact computations modulo powers of five

For this section $p = 5$. To compute a series modulo 5^N, discard a term only after its valuation is known to be at least N. For $x \in 5\mathbb{Z}_5$, Lemma 3.1 gives the uniform bound

$$v_5(x^n/n),\ v_5(x^n/n!)\ \geq \frac{3n+1}{4}. \tag{6.1}$$

In particular, all terms with $n \geq \lceil 4N/3 \rceil$ may be discarded modulo 5^N. This safe cutoff is not always optimal; inspecting individual terms can improve it. The bound concerns denominators as well as powers of x.

6.1. **The logarithm of six.** Here $x = 5$. For precision $5^5 = 3125$, the terms that matter are

$$5, \quad -\frac{25}{2}, \quad \frac{125}{3}, \quad -\frac{625}{4}, \quad \frac{3125}{5} = 625.$$

Their valuations are respectively $1, 2, 3, 4, 4$. The sixth term has valuation 6, and every term with $n \geq 7$ has valuation at least 6 by (6.1). The fifth term must therefore be retained, even though its numerator is divisible by 5^5.

After cancellation, all remaining denominators are invertible modulo 3125. For example, $2^{-1} \equiv 1563$, $3^{-1} \equiv 1042$, and $4^{-1} \equiv 2344 \pmod{3125}$. Thus

$$\begin{aligned} \log(6) &\equiv 5 - 25 \cdot 1563 + 125 \cdot 1042 - 625 \cdot 2344 + 625 \\ &\equiv 1805 \pmod{3125}. \end{aligned}$$

At lower precisions, the same method gives

$$\log(6) \equiv \begin{cases} 55 \pmod{125}, \\ 555 \pmod{625}, \\ 1805 \pmod{3125}. \end{cases} \tag{6.2}$$

These answers are compatible: $1805 \equiv 555 \pmod{625}$ and $555 \equiv 55 \pmod{125}$. Explicitly, at precision 625 one has $-25/2 \equiv 300$ and $125/3 \equiv 250$, whose sum with 5 is 555. At precision 125 only the first two terms contribute.

6.2. **Checking a multiplicative identity.** Since $36 = 6^2$, Theorem 5.1 predicts $\log(36) = 2\log(6)$. We can check the prediction independently by using $36 = 1 + 35$. At precision 625,

$$\log(36) \equiv 35 - \frac{1225}{2} + \frac{42875}{3}.$$

The fourth and fifth terms have valuation 4, the sixth has valuation 6, and (6.1) handles the remaining tail. Using $2^{-1} \equiv 313$ and $3^{-1} \equiv 417 \pmod{625}$ gives

$$-1225/2 \equiv 325, \qquad 42875/3 \equiv 125,$$

so $\log(36) \equiv 485 \pmod{625}$. This agrees with $2 \cdot 555 \equiv 485 \pmod{625}$. It is a check on the computation, not a replacement for the proof of the identity at every precision.

6.3. **The exponential of five.** For $\exp(5)$ modulo 3125, retain terms of degree at most five:

$$\begin{aligned} \exp(5) &\equiv 1 + 5 + \frac{25}{2} + \frac{125}{6} + \frac{625}{24} + \frac{3125}{120} \\ &= 1 + 5 + \frac{25}{2} + \frac{125}{6} + 2\frac{625}{24} \equiv 2956 \pmod{3125}. \end{aligned}$$

The sixth term has valuation $6 - v_5(720) = 5$ and is discarded. All terms of degree at least seven vanish at this precision by (6.1). For an explicit modular check,

$$25/2 \equiv 1575, \quad 125/6 \equiv 2625, \quad 625/24 \equiv 2500 \pmod{3125},$$

so the sum is $1 + 5 + 1575 + 2625 + 5000 \equiv 2956$. Consequently

$$\exp(5) \equiv \begin{cases} 81 \pmod{125}, \\ 456 \pmod{625}, \\ 2956 \pmod{3125}. \end{cases} \tag{6.3}$$

Corollary 5.2 makes inversion stable at finite precision: for example, $\log(456) \equiv 5 \pmod{625}$. A representative of $\exp(5)$ modulo 625 suffices; its unknown higher digits cannot alter this logarithmic congruence.

7. What changes at the prime two

The distinction between $1 + 2\mathbb{Z}_2$ and $1 + 4\mathbb{Z}_2$ is structural, not just a convenience in estimates.

7.1. **Failure of the exponential at the boundary.** By the factorial formula in Lemma 3.1,

$$v_2(2^n/n!) = n - v_2(n!) = s_2(n).$$

For $n = 2^k$, the digit sum is 1. Thus the terms of the exponential series at 2 do not tend to zero, and $\exp(2)$ is undefined by this series. The same argument applies to every input of valuation one. On $4\mathbb{Z}_2$, the estimates and inverse identities already proved do apply.

7.2. **The logarithm on the larger principal-unit group.** The logarithmic series still converges on $1 + 2\mathbb{Z}_2$. We now determine its kernel and image without assuming an exponential inverse there.

Lemma 7.1. *For every $u \in 1 + 2\mathbb{Z}_2$,*

$$\log(u^2) = 2\log(u). \tag{7.1}$$

Proof. Formally, $L(2X + X^2) = 2L(X)$: their derivatives are both $2/(1 + X)$ and their constant terms are zero. To evaluate this at $x = u - 1 \in 2\mathbb{Z}_2$, expand each polynomial $(2x + x^2)^n$. A term of degree m occurs only for $n \le m \le 2n$ and has an integer coefficient divided by n. Its valuation is at least $m - v_2(n) \ge m - \lfloor \log_2 m \rfloor$, which tends to infinity. Only finitely many expanded terms have any given total degree. The rearrangement principle therefore applies to this substitution on the larger domain. This proves the evaluated identity. □

Every odd 2-adic unit u has a unique decomposition

$$u = \varepsilon w, \qquad \varepsilon \in \{1, -1\}, \quad w \in 1 + 4\mathbb{Z}_2 : \tag{7.2}$$

choose $\varepsilon = 1$ if $u \equiv 1 \pmod 4$ and $\varepsilon = -1$ if $u \equiv 3 \pmod 4$. Also $u^2 \in 1 + 8\mathbb{Z}_2$, since for $u = 1 + 2a$, $u^2 - 1 = 4a(a + 1)$ and one of $a, a + 1$ is even.

Proposition 7.2. *The logarithmic series defines a surjective group homomorphism*

$$\log : (1 + 2\mathbb{Z}_2, \times) \longrightarrow (4\mathbb{Z}_2, +),$$

with kernel $\{1, -1\}$. It restricts to the isomorphism on $1 + 4\mathbb{Z}_2$ from Theorem 5.1.

Proof. For the decomposition (7.2), $u^2 = w^2$. Applying Lemma 7.1 to u and to w gives $2\log u = 2\log w$, hence $\log u = \log w$ in the field $\mathbb{Q}_2$. In particular, $\log(-1) = \log(1) = 0$. The decomposition is multiplicative: if $u_j = \varepsilon_j w_j$, then $u_1 u_2 = (\varepsilon_1 \varepsilon_2)(w_1 w_2)$ is the same unique decomposition. The logarithm is additive on products of the w_j by Theorem 5.1, so it is additive on products of the u_j. Its image is exactly $4\mathbb{Z}_2$, and it vanishes exactly when $w = 1$, again by that theorem. This identifies the image and kernel. □

For a concrete first-term comparison, $3 = (-1)(-3)$ and $-3 = 1 - 4$. Thus $\log(3) = \log(1-4)$ has valuation 2, whereas $v_2(3-1) = 1$. The isometry (5.2) would be false on the larger group $1 + 2\mathbb{Z}_2$. The element -1 is nontrivial torsion in that group, while the additive group $2\mathbb{Z}_2$ is torsion-free, so a group isomorphism between them is impossible even without considering series.

8. Extending the logarithm and identifying what it forgets

For odd p, the principal-unit logarithm can be extended to all of $\mathbb{Q}_p^\times$ after choosing a value at p. We construct the convention with value zero at p and compute its kernel.

8.1. **Teichmüller representatives for odd primes.**

Proposition 8.1. *For odd p and each $\overline{a} \in \mathbb{F}_p^\times$, there is a unique $\omega(\overline{a}) \in \mathbb{Z}_p^\times$ such that*

$$\omega(\overline{a}) \equiv \overline{a} \pmod{p}, \qquad \omega(\overline{a})^{p-1} = 1.$$

These representatives form the group μ_{p-1} of $(p-1)$st roots of unity in $\mathbb{Q}_p$. Every unit has a unique factorization

$$u = \omega(\overline{u})\langle u \rangle, \qquad \langle u \rangle \in 1 + p\mathbb{Z}_p. \tag{8.1}$$

Both factors depend multiplicatively on u.

Proof. Let $f(T) = T^{p-1} - 1$. A nonzero residue $\overline{a}$ is a root of f modulo p by Fermat's theorem, and $f'(a) = (p-1)a^{p-2}$ is a unit. Suppose an integer a_n has already been chosen with $f(a_n) \equiv 0$ modulo p^n. The possible next digits have the form $a_{n+1} = a_n + tp^n$ with $t \in \{0, \dots, p-1\}$. Polynomial expansion gives

$$f(a_n + tp^n) \equiv f(a_n) + tp^n f'(a_n) \pmod{p^{n+1}},$$

since all higher terms contain p^{2n} and $2n \geq n+1$. Dividing by p^n leaves a linear congruence for t with invertible coefficient $f'(a_n)$. There is exactly one next digit. The compatible sequence is Cauchy, and its limit is a root with the specified residue. Any root with that residue has the same successive digits, proving uniqueness.

Products of such roots are again roots with the product residue, so uniqueness gives $\omega(\overline{a}\,\overline{b}) = \omega(\overline{a})\omega(\overline{b})$. There is one for each of the $p-1$ nonzero residues, and no further roots: any root is a unit and is determined by its residue. Dividing u by its representative gives (8.1). Its uniqueness and multiplicativity follow from those of ω. □

8.2. **A normalized logarithm on the multiplicative group.**

Every $a \in \mathbb{Q}_p^\times$ for odd p now has a unique decomposition

$$a = p^k \omega w, \qquad k \in \mathbb{Z}, \quad \omega \in \mu_{p-1}, \quad w \in U_1.$$

Define

$$\mathrm{Log}_p(a) = \log w. \tag{8.2}$$

The three factors are multiplicative, so Log_p is a group homomorphism from $\mathbb{Q}_p^\times$ to the additive group $p\mathbb{Z}_p$. It is surjective by Theorem 5.1, and

$$\ker \mathrm{Log}_p = p^{\mathbb{Z}}\mu_{p-1}, \qquad \mathbb{Q}_p^\times / (p^{\mathbb{Z}}\mu_{p-1}) \simeq (p\mathbb{Z}_p, +). \tag{8.3}$$

Indeed, $\mathrm{Log}_p(a) = 0$ if and only if $w = 1$. Continuity follows because the valuation and residue factors are locally constant on $\mathbb{Q}_p^\times$, while the principal-unit logarithm is continuous.

For $p = 2$, use instead the unique factorization $a = 2^k \varepsilon w$ with $\varepsilon \in \{\pm 1\}$ and $w \in 1 + 4\mathbb{Z}_2$. The same definition yields image $4\mathbb{Z}_2$ and kernel $2^{\mathbb{Z}}\{\pm 1\}$. On $1 + 2\mathbb{Z}_2$ it agrees with the series by Proposition 7.2.

Our choice $\mathrm{Log}_p(p) = 0$ is a normalization, not a consequence of (2.1), which does not converge at p as an argument of log. For any $\ell \in \mathbb{Q}_p$ another continuous extension is

$$\mathrm{Log}_{p,\ell}(a) = \mathrm{Log}_p(a) + v_p(a)\ell. \tag{8.4}$$

It agrees on units and takes value ℓ at p. Conversely, an additive-valued group homomorphism agreeing with our principal-unit logarithm kills every root of unity: if $\zeta^m = 1$, then $mh(\zeta) = h(1) = 0$, and the additive group of $\mathbb{Q}_p$ has no torsion. The factorization therefore forces the formula (8.4) once $h(p) = \ell$ is chosen. The kernel assertion (8.3) concerns the normalization $\ell = 0$.

Example 8.2 (A logarithm whose series is not directly available)**.** For $p = 5$, the Teichmüller representative of 2 is 7 modulo 25: $7 \equiv 2 \pmod 5$ and $7^4 = 2401 \equiv 1 \pmod{25}$, and Proposition 8.1 proves uniqueness at this precision. Hence the principal part of 2 is $2/\omega(2) \equiv 2/7 \equiv 11 \pmod{25}$, since $7^{-1} \equiv 18$. It follows that

$$\mathrm{Log}_5(2) \equiv \log(11) \equiv 10 \pmod{25}. \tag{8.5}$$

The ordinary series with $2 = 1 + 1$ does not converge, since its input $x = 1$ has valuation zero. Thus (8.5) uses the extension, not a direct evaluation outside the series domain.

9. Which structure is preserved?

The group isomorphism (5.1) preserves an operation by changing its interpretation:

$$\text{multiplication of principal units} \quad \longmapsto \quad \text{addition in a } p\text{-adic ideal.}$$

It is not a ring homomorphism between the inherited structures. The source U_r is not a ring under the inherited addition and multiplication. Moreover, a unital ring homomorphism would send 1 to 1, whereas $\log(1) = 0$. The product identity here is $\log(uv) = \log u + \log v$, rather than $\log(uv) = \log u \, \log v$.

There is a concrete failure of additivity even for the normalized extension: in $\mathbb{Q}_5$, both arguments $1, 1$ and their sum 2 are nonzero, but

$$\mathrm{Log}_5(1 + 1) = \mathrm{Log}_5(2) \equiv 10 \pmod{25}, \qquad \mathrm{Log}_5(1) + \mathrm{Log}_5(1) = 0.$$

The extension is defined on $\mathbb{Q}_5^\times$, not on an additive subgroup, but this particular sum already rules out an additive extension to a ring homomorphism on $\mathbb{Q}_5$.

The precise amount of information retained depends on the domain. On U_r for $r \geq r_0$, nothing is lost: exponential recovers the original unit. On all of $\mathbb{Q}_p^\times$, the normalized extension forgets the factor p^k and the finite root-of-unity factor. Given only $\mathrm{Log}_p(a)$, one can recover the principal part w, but recovering a additionally requires k and the torsion factor.

One can transport operations through any bijection, but the resulting operation must be stated. For example, on U_r define

$$u \star v = \exp(\log u \, \log v).$$

The product of the two logarithms lies in $p^{2r}\mathbb{Z}_p \subseteq p^r\mathbb{Z}_p$, so this is defined. Together with the original group multiplication as an additive operation, $\star$ gives a nonunital ring isomorphic to the ideal $p^r\mathbb{Z}_p$ with its usual operations. Its additive zero is the unit $1 \in U_r$. Its ring multiplication $\star$ is a new operation; it is not the original multiplication of elements of $\mathbb{Q}_p$. There is no multiplicative identity in this ring: an identity would correspond to an

element $e \in p^r\mathbb{Z}_p$ with $ea = a$ for every $a \in p^r\mathbb{Z}_p$, forcing $e = 1 \notin p^r\mathbb{Z}_p$. This explicitly separates transport of structure from preservation of inherited structure.

10. The limited connection with the log-link in IUT

Mochizuki's overview describes the classical local p-adic logarithm as the non-ring-homomorphic comparison involved in a log-link [1, §3, printed pp. 29–30]. The elementary content developed here is relevant because it makes the domains, operations, convergence conditions, and torsion issues visible.

Three distinctions should remain explicit. First, the isomorphism of principal units with an additive ideal is a local calculation in one fixed completed field. Second, extending the logarithm to a larger multiplicative group introduces a kernel and, on nonunits, a normalization. Third, neither calculation supplies an identification preserving the field's entire ring structure.

The complete IUT log-link is not simply the arrow $1 + p\mathbb{Z}_p \to p\mathbb{Z}_p$ with all other data omitted. The IUT constructions also involve arithmetic fundamental groups, reconstructed structures, and compatibility with other links. We have not constructed those data, proved their compatibility, or established the estimates of IUT. Our results instead provide a rigorous classical component and a set of questions to ask when reading the more elaborate construction: which units are used, which torsion is removed, which topology is retained, and which operations are actually compared?

11. Exercises with complete solutions

Exercise 1. For odd p, prove that U_r is torsion-free for every $r \geq 1$. Explain why $1 + 2\mathbb{Z}_2$ is not torsion-free, but $1 + 4\mathbb{Z}_2$ is.

Solution. If $u \in U_r$ and $u^m = 1$ for a positive integer m, then $m \log u = \log(1) = 0$. The additive group of $\mathbb{Q}_p$ is torsion-free because it is a field of characteristic zero. Thus $\log u = 0$, and Theorem 5.1 implies $u = 1$. The same proof works on $1 + 4\mathbb{Z}_2$. On $1 + 2\mathbb{Z}_2$, however, $-1 = 1 - 2$ belongs to the group, is not 1, and has order two. □

Exercise 2. Compute $\log(456) \pmod{625}$ directly from its series, without first using the inverse identity. Compare with (6.3).

Solution. Put $x = 455 = 5 \cdot 91$. Only degrees one, two, and three contribute: degrees four and five have valuation 4, degree six has valuation 6, and (6.1) handles every higher term. Modulo 625, $x^2 \equiv 150$ and $x^3 \equiv 125$. Thus

$$\log(456) \equiv 455 - 150/2 + 125/3 \equiv 455 - 75 + 250 = 630 \equiv 5 \pmod{625}.$$

This agrees with the fact that 456 represents $\exp(5)$ modulo 625. All denominators in the displayed computation are units modulo 625. □

Exercise 3. For $N \geq 2$, prove that the class of 6 generates $(1+5\mathbb{Z}_5)/(1+5^N\mathbb{Z}_5)$, and determine its order.

Solution. Proposition 3.2 gives $v_5(\log 6) = 1$, so $\log 6 = 5b$ for a unit $b \in \mathbb{Z}_5^\times$. Under Corollary 5.2, its class corresponds to the additive class of $5b$ in $5\mathbb{Z}_5/5^N\mathbb{Z}_5$. Multiplication by the unit b is an automorphism of that additive quotient, and 5 generates it. For a positive integer m, the class of $m\log 6$ is zero precisely when $v_5(m)+1 \geq N$, or $5^{N-1} \mid m$. The order is therefore 5^{N-1}, which is also the size of the quotient. □

Exercise 4. Let p be odd, $r \geq 1$, $m \in \mathbb{Z}\setminus\{0\}$, and $u \in U_r$. Give a necessary and sufficient condition for $v^m = u$ to have a solution $v \in U_r$, and prove uniqueness when a solution exists. Apply the result to $v^5 = 26$ and $v^5 = 6$ in $1+5\mathbb{Z}_5$.

Solution. By the group isomorphism, the equation is equivalent to $m\log v = \log u$ with $\log v \in p^r\mathbb{Z}_p$. It has a solution exactly when

$$\log u \in mp^r\mathbb{Z}_p = p^{r+v_p(m)}\mathbb{Z}_p.$$

If this holds, the unique solution is $v = \exp(\log u/m)$. Uniqueness follows because multiplication by $m \neq 0$ is injective on the additive subgroup of $\mathbb{Q}_p$. For $p=5$, $v_5(\log 26) = v_5(25) = 2$, so $v = \exp(\log 26/5)$ is a unique solution in $1+5\mathbb{Z}_5$. For 6, the logarithm has valuation one, which is less than $r + v_5(5) = 2$, so no such solution exists. This is a statement about solutions in the specified group; it does not rule out roots in an algebraic extension of $\mathbb{Q}_5$. □

Exercise 5. Compute $\mathrm{Log}_5(10)$ and $\mathrm{Log}_5(1/2)$ modulo 25 under the normalization (8.2). Describe all $a \in \mathbb{Q}_5^\times$ with $\mathrm{Log}_5(a) = 0$.

Solution. Since $\mathrm{Log}_5(5) = 0$ and $\mathrm{Log}_5(2) \equiv 10 \pmod{25}$ by (8.5),

$$\mathrm{Log}_5(10) \equiv 10 \pmod{25}, \qquad \mathrm{Log}_5(1/2) \equiv -10 \equiv 15 \pmod{25}.$$

By (8.3), all zero-logarithm elements, and only those, have the form $5^k\zeta$, with $k \in \mathbb{Z}$ and $\zeta^4 = 1$ in $\mathbb{Q}_5$. The four possibilities for ζ are the Teichmüller lifts of the nonzero residues modulo 5. A nonzero element with logarithm zero need not be the number 1; 5 is already a counterexample. □

Exercise 6. For odd p and $r \geq 1$, let $u \in U_r$ with $u \neq 1$. Compare its original product u^2 with the transported product $u \star u = \exp((\log u)^2)$. Show that they are never equal.

Solution. Put $a = \log u$. Theorem 5.1 gives $a \neq 0$ and $a \in p^r\mathbb{Z}_p$. If $u^2 = u \star u$, applying logarithm yields $2a = a^2$, hence $a(a-2) = 0$ in the field $\mathbb{Q}_p$. Thus $a = 2$, since $a \neq 0$. For odd p, the element 2 is a unit, so it cannot lie in $p^r\mathbb{Z}_p$. This contradiction shows that the two products are different. The transported nonunital ring is perfectly well-defined, but its multiplication is not the field multiplication restricted to principal units. □

References

[1] S. Mochizuki, *A Panoramic Overview of Inter-universal Teichmüller Theory*, RIMS Kôkyûroku Bessatsu **B51** (2014), 301–345, especially §3, printed pp. 29–30. https://www.kurims.kyoto-u.ac.jp/~motizuki/Panoramic%20Overview%20of%20Inter-universal%20Teichmuller%20Theory.pdf.